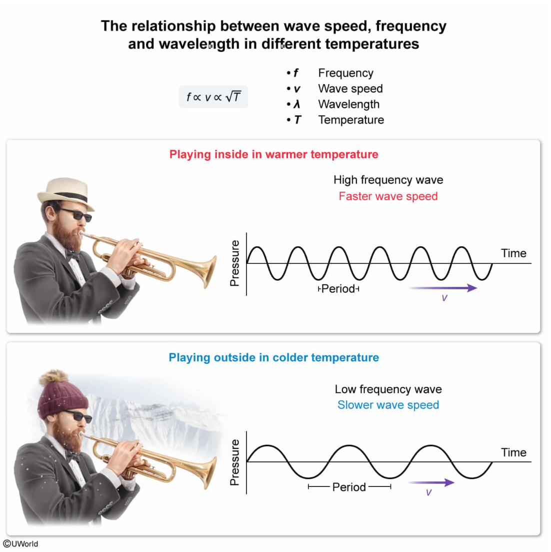
A musician plays the trumpet in a marching band. The musician tunes the trumpet inside a room and then goes outside to play. When outside, the trumpet sounds flat (plays a lower frequency). Which of the following has occurred to account for the decrease in the frequency?

- A. The air outside is warmer than inside, which increased the speed of the sound waves. [7%]
- B. More energy is dissipated outside, which caused the sound waves to lose energy. [34%]
- C. The musician shortened the length of the trumpet, which decreased the wavelength of the sound waves. [3%]
- D. The air outside is cooler than inside, which decreased the speed of the sound waves. [55%]

Correct
54% answered correctly

Explanation:



A **wave** is a motion created by a disturbance in a medium. A wave moves through the medium with a speed v that is related to the wavelength λ and frequency f:

$$v = \lambda \ f$$

Rearranging, the frequency of a sound wave is directly proportional to v and inversely proportional to λ:

$$f = \frac{v}{\lambda}$$

The speed of waves is determined by the physical properties of the **medium** through which they travel. For example, the speed of sound waves depends on the **temperature** of the medium: sound travels faster in warmer air and slower in cooler air. In particular, the speed of sound in air follows the following formula (in kelvin, K):

$$v = 331\,\frac{\mathrm{m}}{\mathrm{s}} \cdot \sqrt{\frac{T}{273\ \mathrm{K}}}$$

Consequently, when the wavelength is constant, an increase (or decrease) in wave speed results in a corresponding increase (or decrease) in the wave's frequency.

In this question, the frequency of the trumpet's sound is flat when the musician plays outside. Because the wavelength of the sound is fixed by the size of the tube, the decrease in frequency is caused by a decrease in v. Therefore, the decrease in frequency can be accounted for by the air outside being cooler than the air inside, which decreases the speed of the sound waves.

(Choice A) Sound waves travel faster in warmer air. Hence, the speed of sound waves increases, causing the frequency to increase, not decrease.

(Choice B) The energy of a wave is related to the wave's intensity and loudness but not the wave's frequency. Dissipation of energy does not affect the frequency of a wave.

(Choice C) The wavelength of a sound wave produced by a brass instrument depends on the length of the tube. A shorter trumpet results in a shorter wavelength and a *higher* frequency.

Educational objective:
The frequency of sound is directly related to wave speed and inversely related to wavelength. Sound waves travel faster in warmer air and slower in cooler air.

Time spent: 0 sec	**QID**:403982
Subject: Physics	**System:** Light and Sound

Researchers studied the effects of glaucoma by measuring the pressure inside the eye, called the intraocular pressure, and testing for vision abnormalities in different groups of patients, as shown in Table 1. The intraocular pressure is maintained by the constant generation and drainage of the fluid within the eye, the vitreous humor. When the drainage pathway becomes blocked, the intraocular pressure increases and glaucoma can develop.

Table 1 Effects of Different Stages of Glaucoma

Patient Group	Intraocular pressure (N/m^2)	Vision abnormalities
Normal	2,000	None
Early glaucoma	3,000	None
Moderate glaucoma	4,000	Moderate peripheral loss

Glaucoma and other eye disorders are often treated with laser surgery because lasers generate high-intensity light within a very narrow band of frequencies. This feature allows specific treatment of different tissues based on the type of laser used. For example, cataract surgery uses a laser that passes through the cornea with its energy absorbed in the lens. The frequency of light and surgical application for some common medical lasers are listed in Table 2.

Table 2 Lasers Used for Eye Surgeries

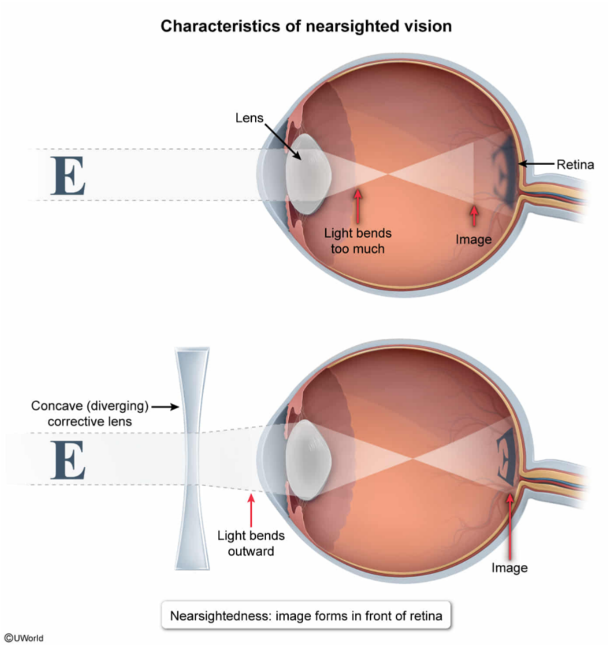
Laser	Frequency (THz)	Application
Argon	500	Cornea surgery
Diode	370	Retina surgery
Krypton red	460	Retina surgery
Nd-YAG	560	Glaucoma surgery

Which of the following is NOT a characteristic of nearsighted vision?

- A. Eye lens bends light too much [13%]
- B. Corrected with a diverging lens [20%]
- C. Corrected with a concave lens [22%]
- ✔ D. Image forms behind the retina [43%]

Correct
43% answered correctly

Explanation:



Light entering the eye is refracted, or bent, by the lens to form an image on the retina at the back of the eye. Vision problems occur when the lens does not correctly refract the light entering the eye and the image forms in front of or behind the retina.

In this question, **nearsightedness** (also called myopia) occurs when the lens bends light too much and the image forms in front of the retina. Nearsighted vision is corrected by spreading out the light rays before they are refracted by the lens (ie, making the light rays diverge), which is accomplished with a concave or diverging lens. Therefore, the only listed characteristic not associated with nearsightedness is the image forming behind the retina.

(Choices A, B, and C) Nearsightedness occurs when the lens bends light too much, and it is corrected with a concave lens (also called a diverging lens).

Educational objective:
In nearsighted vision, the lens of the eye bends light too much, forming the image in front of the retina. It is corrected using a concave lens, also known as a diverging lens.

Time spent: 0 sec	**QID:** 403983
Subject: Physics	**System:** Light and Sound

Researchers studied the effects of glaucoma by measuring the pressure inside the eye, called the intraocular pressure, and testing for vision abnormalities in different groups of patients, as shown in Table 1. The intraocular pressure is maintained by the constant generation and drainage of the fluid within the eye, the vitreous humor. When the drainage pathway becomes blocked, the intraocular pressure increases and glaucoma can develop.

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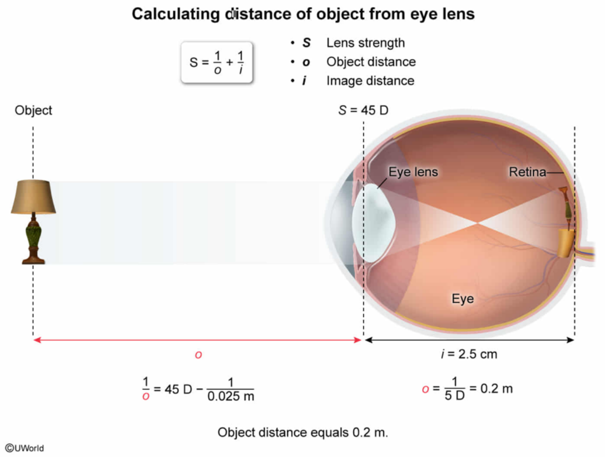
A patient with a lens-to-retina distance of 2.5 cm and a lens strength of 45 D can clearly see an object. What is the distance from the patient's eye to the object?

- A. 0.2 m [31%]
- B. 0.3 m [26%]
- C. 1 m [25%]
- D. 5 m [16%]

Correct
30% answered correctly

Explanation:



Lens strength *S* is equal to the inverse of the focal length *f* of the lens:

$$S = \frac{1}{f}$$

The **thin lens equation** is used to calculate *S* in terms of the distance from the lens to the object *o* and to the image *i*:

$$S = \frac{1}{o} + \frac{1}{i}$$

The units for *S* are **diopters** (abbreviated D), which are equivalent to m^{-1}.

In this question, *S* equals 45 D and *i* equals 2.5 cm, which is equal to 0.025 m. Substituting these values into the above equation gives:

$$45\text{ D} = \frac{1}{o} + \frac{1}{0.025\text{ m}}$$

$$45\text{ D} = \frac{1}{o} + 40\text{ D}$$

Solving for *o* gives:

$$\frac{1}{o} = 45\text{ D} - 40\text{ D} = 5\text{ D}$$

$$o = \frac{1}{5\text{ D}} = 0.2\text{ m}$$

Therefore, the distance *o* equals 0.2 m.

(Choice B) A distance of 0.3 m results from summing the inverse of the lens strength and the image distance instead of using the thin lens equation.

(Choice C) An object distance of 1 m is calculated from the product of the lens strength and the image distance.

(Choice D) A value of 5 m is obtained by assuming the inverse of the object distance (5 D) equals the object distance.

Educational objective:

Lens strength equals the inverse of the focal length of the lens. The thin lens equation calculates lens strength from the lens-to-object distance and the lens-to-image distance.

Time spent:0 sec	**QID**:403984
Subject:	**System:**
Physics	Light and Sound

Researchers studied the effects of glaucoma by measuring the pressure inside the eye, called the intraocular pressure, and testing for vision abnormalities in different groups of patients, as shown in Table 1. The intraocular pressure is maintained by the constant generation and drainage of the fluid within the eye, the vitreous humor. When the drainage pathway becomes blocked, the intraocular pressure increases and glaucoma can develop.

Table 1 Effects of Different Stages of Glaucoma

Patient Group	Intraocular pressure (N/m^2)	Vision abnormalities
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Moderate glaucoma	4,000	Moderate peripheral loss

Glaucoma and other eye disorders are often treated with laser surgery because lasers generate high-intensity light within a very narrow band of frequencies. This feature allows specific treatment of different tissues based on the type of laser used. For example, cataract surgery uses a laser that passes through the cornea with its energy absorbed in the lens. The frequency of light and surgical application for some common medical lasers are listed in Table 2.

Table 2 Lasers Used for Eye Surgeries

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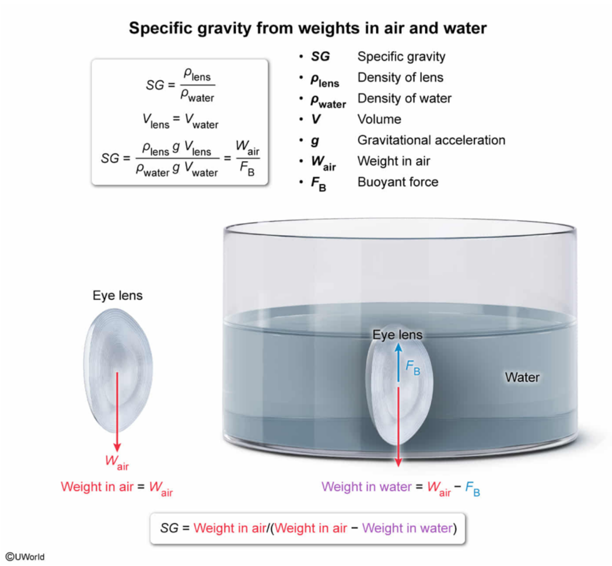
Nd-YAG	560	Glaucoma surgery

A researcher is measuring the specific gravity of the lens from a patient's eye. If the weight of the lens in air is known, what additional information is needed to determine the specific gravity of the lens?

- A. Focal length of the lens [8%]
- ✔ B. Weight of the lens submerged in water [56%]
- C. Volume of water in the lens [29%]
- D. Speed of light in the lens [5%]

Correct
57% answered correctly

Explanation:



The **density** ρ of a material equals the ratio of its mass m and volume V:

$$\rho = \frac{m}{V}$$

Furthermore, an object's **specific gravity** SG is the ratio of its density ρ_{obj} to the density of water (ρ_{water} = 1 kg/L):

$$\mathrm{SG} = \frac{\rho_{\mathrm{obj}}}{\rho_{\mathrm{water}}}$$

When an object is submerged in water, it displaces a volume V of water equal to its own volume (ie, $V_{obj} = V_W$). Multiplying the numerator and denominator of the equation above by the product of V and gravitational acceleration g yields:

$$\mathrm{SG} = \frac{\rho_{\mathrm{obj}}\mathrm{gV}}{\rho_{\mathrm{W}}\mathrm{gV}}$$

The numerator of this equation equals the weight of the object in air, $W_{obj.air}$:

$$W_{\mathrm{obj,air}} = \rho_{\mathrm{obj}}\mathrm{gV}$$

and the denominator equals the **buoyant force** F_B acting on the object when it is submerged in water:

$$F_{\mathrm{B}} = \rho_{\mathrm{W}}\mathrm{gV}$$

Hence:

$$\mathrm{SG} = \frac{W_{\mathrm{obj,air}}}{F_{\mathrm{B}}}$$

In this question, the weight of the lens in air W_{air} is known, and the apparent weight of the lens submerged in water W_{water} equals the difference between W_{air} and F_B:

$$W_{\text{water}} = W_{\text{air}} - F_{\text{B}}$$

Therefore, the ratio of W_{air} and the difference of W_{air} and W_{water} is equal to the *SG* of the lens:

$$\frac{W_{\text{air}}}{W_{\text{air}} - W_{\text{water}}} = \frac{W_{\text{air}}}{W_{\text{air}} - (W_{\text{air}} - F_{\text{B}})} = \frac{W_{\text{air}}}{\cancel{W_{\text{air}}} - \cancel{W_{\text{air}}} + F_{\text{B}}} = \frac{W_{\text{air}}}{F_{\text{B}}}$$

(Choice A) The focal length of the lens depends on how much it bends light, and does not depend on the specific gravity of the lens.

(Choice C) The volume of water could affect the specific gravity of the lens, but the specific gravity cannot be determined with this information and the weight of the lens in air.

(Choice D) The speed of light in the lens is used to calculate the refractive index of the lens, not the specific gravity.

Educational objective:
The specific gravity of an object equals the ratio of its density to the density of water. This value can be calculated from an object's weights in air and in water using the equation: specific gravity = weight in air/(weight in air − weight in water).

Time spent: 0 sec | **QID:** 403985
Subject: Physics | **System:** Light and Sound

Researchers studied the effects of glaucoma by measuring the pressure inside the eye, called the intraocular pressure, and testing for vision abnormalities in different groups of patients, as shown in Table 1. The intraocular pressure is maintained by the constant generation and drainage of the fluid within the eye, the vitreous humor. When the drainage pathway becomes blocked, the intraocular pressure increases and glaucoma can develop.

Table 1 Effects of Different Stages of Glaucoma

Patient Group	Intraocular pressure (N/m^2)	Vision abnormalities
Normal	2,000	None
Early glaucoma	3,000	None
Moderate glaucoma	4,000	Moderate peripheral loss

Glaucoma and other eye disorders are often treated with laser surgery because lasers generate high-intensity light within a very narrow band of frequencies. This feature allows specific treatment of different tissues based on the type of laser used. For example, cataract surgery uses a laser that passes through the cornea with its energy absorbed in the lens. The frequency of light and surgical application for some common medical lasers are listed in Table 2.

Table 2 Lasers Used for Eye Surgeries

Laser	Frequency (THz)	Application
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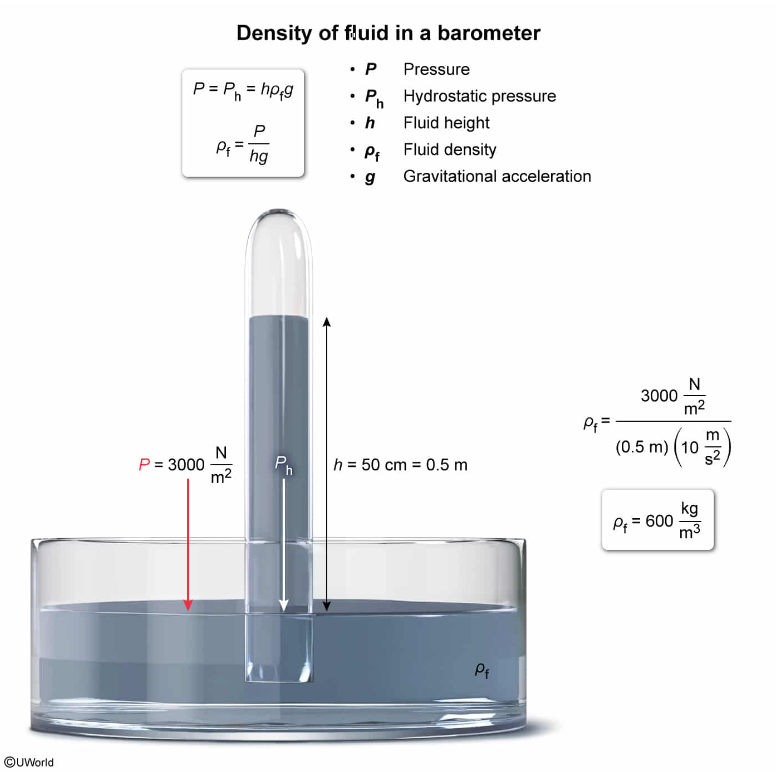
Krypton red	460	Retina surgery
Nd-YAG	560	Glaucoma surgery

When exposed to the intraocular pressure of a patient with early glaucoma, the fluid in a barometer rises to a height of 50 cm. What is the density of the fluid in the barometer?

- A. 1,000 kg/m^3 [8%]
- B. 800 kg/m^3 [15%]
- C. 600 kg/m^3 [71%]
- D. 400 kg/m^3 [4%]

Correct
72% answered correctly

Explanation:



A fluid barometer consists of an evacuated tube with its open end in a fluid-filled container exposed to a pressure P. The fluid rises in the tube until the hydrostatic pressure P_h of the fluid column equals P:

$$P = P_h$$

P_h equals the product of the **height** h of the fluid, the **density** ρ_f of the fluid, and gravitational acceleration g:

$$P = P_h = h\rho_f g$$

Solving the above equation for ρ_f yields:

$$\rho_f = \frac{P}{hg}$$

In this question, a barometer is exposed to a P of 3,000 N/m^2, corresponding to a patient with early glaucoma as shown in Table 1, and the fluid column rises to 50 cm. Substituting these values into the above equation yields:

$$\rho_f = \frac{3{,}000\frac{\text{N}}{\text{m}^2}}{(50\text{ cm})\left(10\frac{\text{m}}{\text{s}^2}\right)}$$

Converting newtons and centimeters to standard units gives:

$$\rho_f = \frac{3{,}000\frac{\text{kg}\cdot\cancel{\text{m}}}{\cancel{\text{s}^2}}\frac{1}{\text{m}^2}}{(0.5\text{ m})\left(10\frac{\cancel{\text{m}}}{\cancel{\text{s}^2}}\right)} = 600\frac{\text{kg}}{\text{m}^3}$$

Therefore, ρ_f equals 600 kg/m^3.

(Choice A) 1,000 kg/m^3 is the density of water, but the barometer is not filled with water.

(Choice B) 800 kg/m^3 is calculated using the intraocular pressure of a moderate glaucoma patient (4,000 N/m^2) instead of the pressure for a patient with early glaucoma.

(Choice D) 400 kg/m^3 results from using the normal intraocular pressure (2,000 N/m^2) instead of the pressure for a patient with early glaucoma.

Educational objective:

A simple fluid barometer can be used to measure pressure, such as atmospheric pressure. The hydrostatic pressure in the barometer's fluid-filled column is equal to the applied pressure.

Time spent:0 sec
Subject:
Physics

QID:403986
System:
Light and Sound

Researchers studied the effects of glaucoma by measuring the pressure inside the eye, called the intraocular pressure, and testing for vision abnormalities in different groups of patients, as shown in Table 1. The intraocular pressure is maintained by the constant generation and drainage of the fluid within the eye, the vitreous humor. When the drainage pathway becomes blocked, the intraocular pressure increases and glaucoma can develop.

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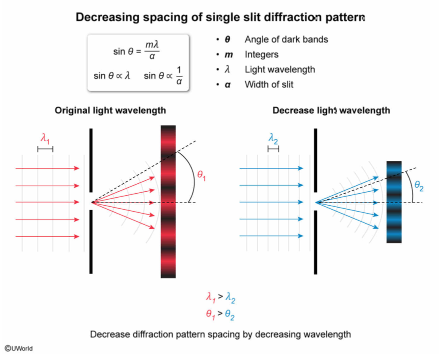
A researcher uses a single slit diffraction pattern to evaluate a patient's vision. Which of the following changes will decrease the spacing of the diffraction pattern?

- A. Decreasing the width of the slit [27%]
- B. Decreasing the wavelength of the light [39%]
- C. Increasing the wavelength of the light [28%]
- D. Increasing the intensity of the light [4%]

Correct
40% answered correctly

Explanation:



Diffraction refers to the bending of light around edges or objects. Light passing through a single slit forms a diffraction pattern of alternating bright and dark bands when the width *a* of the slit is comparable to the wavelength λ of the light.

The waves of light exiting the slit travel different distances based on their original position within the slit. When the distance traveled by two light waves differs by one half of λ, the waves are 180° out of phase and cause destructive interference, forming a dark band. Conversely, a difference in distance traveled equal to an integer multiple m of λ results in constructive interference, forming a bright band in the diffraction pattern.

The dark bands of the diffraction pattern occur at the angles θ relative to the slit, given by the equation:

$$\sin\theta \ = \frac{m\lambda}{a}$$

In this question, the researcher wants to decrease the spacing of the diffraction pattern, so the value of θ should be decreased. Because the sine function increases from 0° to 90°, sin θ decreases when θ decreases. Furthermore, sin θ is directly proportional to λ and inversely proportional to a, implying that θ decreases when λ decreases or when a increases.

Therefore, the researcher can decrease the spacing of the diffraction pattern by decreasing λ or increasing a.

(Choice A) Decreasing the slit width causes the spacing of the diffraction pattern to increase, not decrease.

(Choice C) Increasing the wavelength of the light causes the spacing of the diffraction pattern to increase, not decrease.

(Choice D) Changing the intensity of the light does not affect the spacing of the diffraction pattern.

Educational objective:
Light passing through a single slit generates a diffraction pattern of alternating bright and dark bands due to constructive and destructive interference of the light waves. The spacing of the band pattern depends on the slit width and wavelength of the light.

Time spent: 0 sec
Subject: Physics

QID: 403987
System: Light and Sound

Researchers studied the effects of glaucoma by measuring the pressure inside the eye, called the intraocular pressure, and testing for vision abnormalities in different groups of patients, as shown in Table 1. The intraocular pressure is maintained by the constant generation and drainage of the fluid within the eye, the vitreous humor. When the drainage pathway becomes blocked, the intraocular pressure increases and glaucoma can develop.

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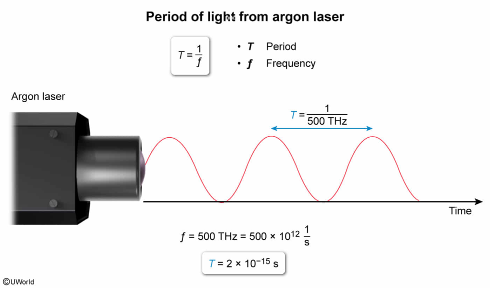
What is the period of the light generated by an argon laser?

- A. 5×10^{-12} s [16%]
- B. 2×10^{-12} s [23%]
- C. 5×10^{-14} s [20%]
- ✔ D. 2×10^{-15} s [39%]

Correct
38% answered correctly

Explanation:



Light is an electromagnetic wave. Waves are characterized by several properties, including **frequency**, period, wavelength, and wave speed. The period T of a wave equals the inverse of its frequency f:

$$T = \frac{1}{f}$$

In this question, an argon laser generates light with f equal to 500 THz, as shown in Table 2. Converting the units of f from THz to 1/s yields:

$$f = 500\ \text{THz} = 500 \times 10^{12}\ \text{Hz} = 5 \times 10^{14}\ \frac{1}{\text{s}}$$

Therefore, T is given by:

$$T = \frac{1}{5 \times 10^{14} \frac{1}{\text{s}}} = 0.2 \times 10^{-14}\ \text{s} = 2 \times 10^{-15}\ \text{s}$$

(Choice A) 5×10^{-12} s corresponds to a frequency of 0.2 THz, but the argon laser has a frequency of 500 THz.

(Choice B) 2×10^{-12} s is calculated by using a factor of 10^9 when converting from THz to Hz instead of 10^{12}.

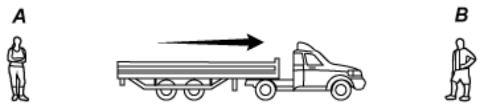
(Choice C) 5×10^{-14} s results from incorrectly calculating the inverse of the frequency.

Educational objective:
The period of a wave is the time between successive wave features (eg, two adjacent peaks or two adjacent troughs). The period equals the inverse of the frequency of the wave.

Time spent:0 sec
Subject:
Physics

QID:403988
System:
Light and Sound

A truck is moving to the right as shown.

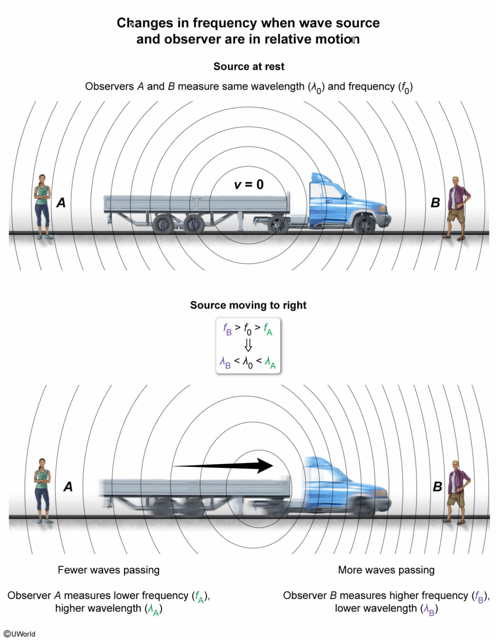


The driver is sounding the horn continually. Which statement about the frequency of the sound heard by observer *A* and observer *B* is true?

- A. Observer *A* hears a higher frequency than observer *B*, because the sound travels slower to reach *A*. [2%]
- B. Observer *B* hears a higher frequency than observer *A*, because the waves that reach observer *B* have a shorter wavelength than those that reach observer *A*. [91%]
- C. Both observers hear the same frequency, because the speed of sound is constant in any given medium. [4%]
- D. Both observers hear the same frequency, because the increase in the speed of sound at observer *B* is offset by an increase in wavelength. [1%]

Correct
91% answered correctly

Explanation:



Two fundamental properties of a wave are its **frequency** *f*, which measures the number of oscillation cycles in a period of time, and its **wavelength** λ, which is the distance between two neighboring points with the same behavior (eg, two crests or two troughs). The wave speed *v* varies depending on the type of medium but is always equal to the product of the frequency and wavelength:

$$v = \lambda \ f$$

When the source of waves (eg, sound created by speakers) and the observer receiving them (eg, human ear) are in motion relative to one another, there is a shift in frequency known as the Doppler effect. For example, when a source creating waves of frequency f_0 moves *toward* a stationary observer, the observed wave frequency f_{obs} is greater than f_0. Conversely, when the source is moving *away*, f_{obs} is less than f_0. Because *v* is constant in a given medium, the corresponding wavelength is inversely proportional to frequency:

$$\lambda = \frac{v}{f}$$

Consequently, the following relationships exist between the observed wavelength λ_{obs}, the stationary wavelength λ_0, f_{obs}, and f_0:

	Frequency	Wavelength
Source moving toward observer	$f_{obs} > f_0$	$\lambda_{obs} < \lambda_0$
Source moving away from observer	$f_{obs} < f_0$	$\lambda_{obs} > \lambda_0$

In this question, the truck driver sounds a horn continually while moving toward observer *B* but away from observer *A*. Therefore, observer *B* hears a higher frequency than observer *A* because the waves reaching observer *B* have a shorter wavelength than those reaching observer *A*.

(Choice A) Observer *A* hears a lower frequency because the wavelength for observer *A* is longer. The sound waves travel the same speed in all directions.

(Choices C and D) Wave speed is the same in all directions, but the wavelength is longer for observer *A* and therefore produces a lower frequency.

Educational objective:
A frequency shift occurs when there is relative motion between the source of a wave and an observer. When the source approaches the observer, the frequency is higher; when the source recedes from the observer, the frequency is lower.

Time spent: 0 sec
Subject: Physics

QID: 403989
System: Light and Sound